

BRAC UNIVERSITY (SET-A)

CSE110 Exam: Quiz 02

Semester: Spring26

Full Marks: 10

Duration: 20 Mins

Name:	ID:	Section:
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Question 1 [10 Points]

Write a Java program that will take two Strings of the same length and an operator character (* or /) from the user. Your task is to create two numbers from the two input strings by considering only the odd digits and perform mathematical operations based on the third input (operator). Finally, print the newly created two numbers and the output value. The input String can contain other characters than digits. For simplicity, assume that both input strings will always be of the same length. You are not allowed to use any built-in String methods other than .length() and .charAt() methods.

Sample Input	Sample Output
ab4s3F67h%21 mnP543&1b80Q *	371 531 197001
5dQ!87 26f0&3 /	57 3 19
Explanation: For sample Input 1, ab4s3F67h%21 contains odd digits 3,7,1, that's why the first number is 371 mnP543&1b86Q contains odd digits 5,3,1, that's why the second number is 531 After performing the addition operation (*), the final output is 197001	

Solution of Set A:

import java.util.Scanner;		
public class quiz{	class	[0.5 marks]
public static void main(String [] args){		
Scanner sc= new Scanner(System.in);	2 input	[0.5 marks]
String string1=sc.next();		
String string2=sc.next();		
String num1="";		
for(int i=0;i<string1.length();i++){	odd digit concat->String 1	[2 marks]
int ascii = string1.charAt(i);		
if((ascii>48&&ascii<=57)&&(ascii%2!=0)){		
num1+=string1.charAt(i);		
}		
}		
String num2="";		
for(int i=0;i<string2.length();i++){	odd digit concat->String 2	[2 marks]
int ascii = string2.charAt(i); //i=0 ascii=97		
if((ascii>48&&ascii<=57)&&(ascii%2!=0)){		
num2+=string2.charAt(i);		
}		
}		
int power= num1.length()-1;	string 1->number 1	[2 marks]
int new_num=0;		
for(int i=0;i<num1.length();i++){		
int ascii=num1.charAt(i);		
int num=ascii-48;		
new_num+=num* Math.pow(10,power);		
power--;		
}		
int power2= num2.length()-1;	string 2->number 2	[2 marks]
int new_num2=0;		
for(int i=0;i<num2.length();i++){		
int ascii=num2.charAt(i);		
int num=ascii-48;//num=1		
new_num2+=num* Math.pow(10,power2);		
power2--;		
}		
String operator=sc.next();	* or /	[1 mark]
if(operator.equals("")){		
System.out.println(new_num*new_num2);		
}		
else{		
System.out.println(new_num/new_num2);		
}		
}}		

BRAC UNIVERSITY (SET-B)

CSE110 Exam: Quiz 02

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Name:	ID:	Section:
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Question 1 [10 Points]

Write a Java program that will take two Strings of the same length and an operator character (+ or -) from the user. Your task is to create two numbers from the two input strings by considering only the odd digits and perform mathematical operations based on the third input (operator). Finally, print the newly created two numbers and the output value. The input String can contain other characters than digits. For simplicity, assume that both input strings will always be of the same length. You are not allowed to use any built-in String methods other than `.length()` and `.charAt()` methods.

Sample Input	Sample Output
ab4s3F67h%21 mnP543&1b80Q +	371 531 902
5dQ!87 26f0&3 -	57 3 54
Explanation: For sample Input 1, ab4s3F67h%21 contains odd digits 3,7,1, that's why the first number is 371 mnP543&1b86Q contains odd digits 5,3,1, that's why the second number is 531 After performing the addition operation (+), the final output is 902	

Solution of Set B:

import java.util.Scanner;		
public class quiz{	class	[0.5 marks]
public static void main(String [] args){		
Scanner sc= new Scanner(System.in);	2 input	[0.5 marks]
String string1=sc.next();		
String string2=sc.next();		
String num1="";		
for(int i=0;i<string1.length();i++){	odd digit concat->String 1	[2 marks]
int ascii = string1.charAt(i);		
if((ascii>48&&ascii<=57)&&(ascii%2!=0)){		
num1+=string1.charAt(i);		
}		
}		
String num2="";		
for(int i=0;i<string2.length();i++){	odd digit concat->String 2	[2 marks]
int ascii = string2.charAt(i); //i=0 ascii=97		
if((ascii>48&&ascii<=57)&&(ascii%2!=0)){		
num2+=string2.charAt(i);		
}		
}		
int power= num1.length()-1;	string 1->number 1	[2 marks]
int new_num=0;		
for(int i=0;i<num1.length();i++){		
int ascii=num1.charAt(i);		
int num=ascii-48;		
new_num+=num* Math.pow(10,power);		
power--;		
}		
int power2= num2.length()-1;	string 2->number 2	[2 marks]
int new_num2=0;		
for(int i=0;i<num2.length();i++){		
int ascii=num2.charAt(i);		
int num=ascii-48;//num=1		
new_num2+=num* Math.pow(10,power2);		
power2--;		
}		
String operator=sc.next();	* or /	[1 mark]
if(operator.equals("+")){		
System.out.println(new_num+new_num2);		
}		
else{		
System.out.println(new_num-new_num2);		
}		
}}		